

Name: _____ Date: _____ Period: _____

Warm Up - Convert the following rational numbers to the other two forms.

1. $\frac{4}{5} = 0.8, 80\%$

2. $45\% = 0.45$

3. $0.375 = \frac{3}{8}, 37.5\%$

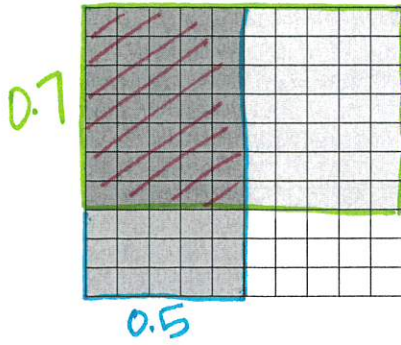
$\frac{4}{5} \cdot \frac{2}{2} = \frac{8}{10} = 0.80$
D2P

$\frac{45}{100} = \frac{9}{20}$

$\frac{375}{1000} \div 125 = \frac{3}{8}$

NOTES: Multiplying with Decimals

Modeling

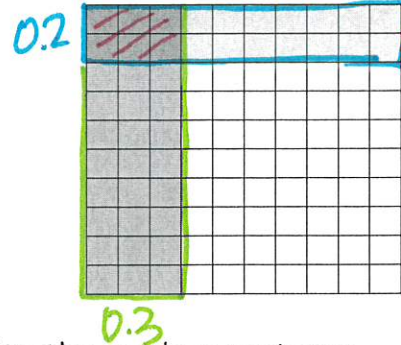


1. Write the number sentence.

$(0.5)(0.7) = 0.35$

What is the problem "saying"?

Half of seven-tenths



2. Write the number sentence.

$(0.3)(0.2) = 0.06$

What is the problem "saying"?

Part of part

True or False: Multiplying always makes the product bigger.

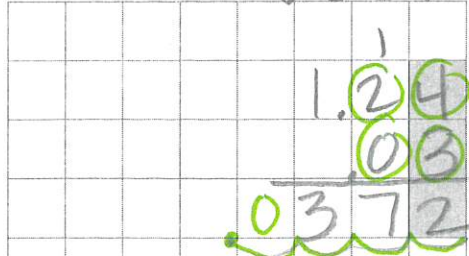
Does the quantity increase or decrease when multiplying by a decimal less than one? Decrease Why? You're finding part of a part

Steps to Solve:

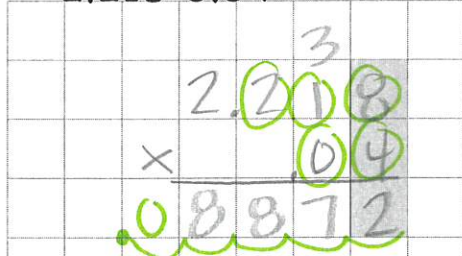
1. Ignore the decimals and multiply the numbers.
2. Count the numbers behind the decimals.
3. Start from the right. Count that many spaces over.

Try on your own. Do NOT line up decimals. Do NOT add zeroes

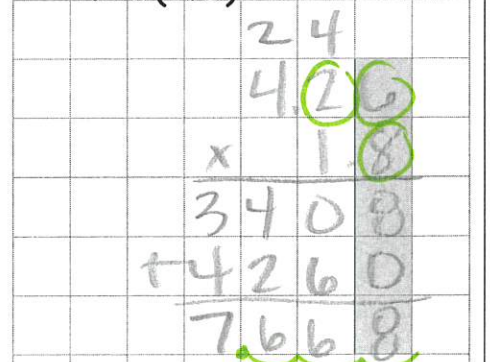
$1.24(0.03) = 0.0372$



$2.218 \cdot 0.04 = 0.08872$



$4.26(1.8) = 7.668$



Multiplying Decimals

Michelle had planned to run a marathon which is 26.1 miles long. She had only run 0.7 of that before she twisted her ankle. How many miles did she run before she got hurt?

18.27 miles

$$\begin{array}{r} 26.1 \\ \times 0.7 \\ \hline 18.27 \end{array}$$

$$4.9 (0.02) = 0.098$$

$$\begin{array}{r} 4.9 \\ \times 0.02 \\ \hline 0.098 \end{array}$$

$$0.56 \cdot 0.03 = 0.0168$$

$$\begin{array}{r} 0.56 \\ \times 0.03 \\ \hline 0.0168 \end{array}$$

$$4.68 (0.034) = 0.15912$$

$$\begin{array}{r} 4.68 \\ \times 0.034 \\ \hline 18.72 \\ + 140.40 \\ \hline 159.12 \end{array}$$

Ishi makes \$8.50 an hour rolling sushi at his new job. When he receives his paycheck he notices he worked 20.88 hours. How much does Ishi make before taxes?

\$177.48

$$\begin{array}{r} 20.88 \\ \times 8.50 \\ \hline 10440 \\ + 167040 \\ \hline 177.480 \end{array}$$

Ms. Morgan bought 3.5 pounds of bananas at \$0.51 a pound and 4.5 pounds of pineapple at \$1.19 a pound. How much did she pay for each?

$$\begin{array}{r} 3.5 \\ \times 0.51 \\ \hline 35 \\ 1750 \\ \hline 1.785 \end{array} \quad \begin{array}{r} 1.19 \\ \times 4.5 \\ \hline 595 \\ 4760 \\ \hline 5.355 \end{array}$$

\$1.79

\$5.36